Intelligent Systems

Constraints Satisfaction Problem

Quick Recap of Last Class

- Introduction to CSP
- Types of CSP
- Backtracking Search

In Today’s class

- Heuristics for CSP
- Forward Checking

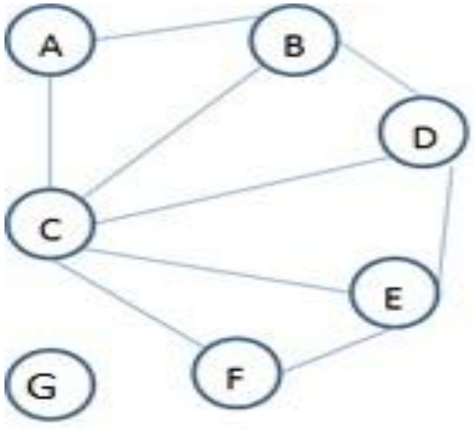
Improving backtracking efficiency

- Plain backtracking is not efficient. We can get considerable performance improvement if we answer the following questions

  - ***Which variable should be assigned next?***
  - ***In what order should its values be tried?***

Heuristics in CSP

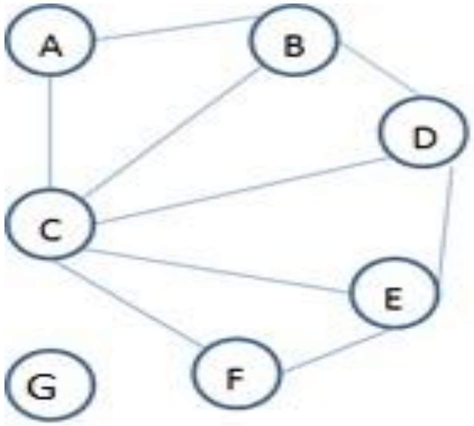
- Domain={1,2,3}
- Suppose A = 1 and B = 2
  - The only choice left with C is 3, whereas D has two choices.
  - It make sense to choose variable C and assign C = 3.



- This is called as ***Minimum Remaining Value (MRV)*** heuristic.
- *MRV heuristics says choose a variable with the "fewest" legal values.*

Heuristics in CSP

- Domain={1,2,3}
- Suppose A = 1 and B = 2
    - Suppose next variable chosen is D.
    - Shall we assign D = 3 or D = 1?
    - If D = 3 then no choice would be left for variable C. Therefore choose D=1

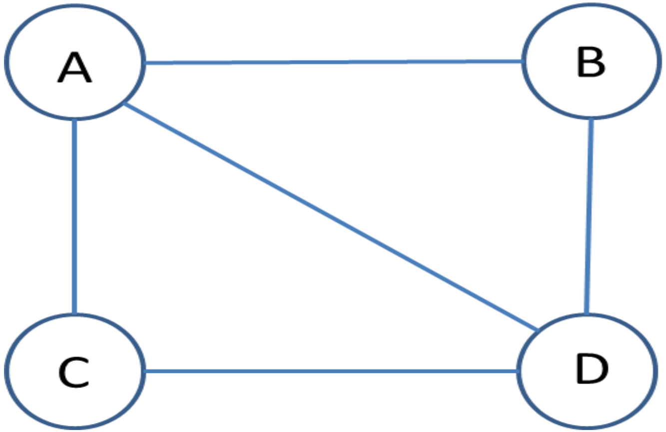


- This is called as ***Least Constraining Value (LCV)*** heuristic.
- *LCV heuristics says choose a value for the current node that leaves more number of choices open for neighboring nodes.*

Heuristics in CSP

- ***Degree Heuristic***
  - MRV heuristic will not be effective at the start at which all the variables have equal number of remaining values. Therefore use degree heuristic.
- *Degree heuristic says choose a variable with highest degree.*

Example



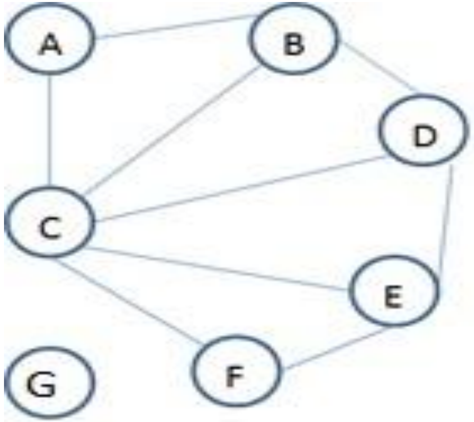
Heuristics in CSP are problem independent

- If a problem is treated as CSP, the main advantage we get is we can develop problem independent heuristics.

Forward Checking

- The idea
  - So far our search considers the constraints on a variable only at the time that variable is chosen for assigning value
  - But by looking at some of the constraints earlier in the search, we can drastically reduce the search space.
- ***Forward Checking***
  - Cross-off values that violate a constraint when added to the existing assignment

Example: Forward Checking



Thank you